

Manuel David MAYA ROSERO
AI Software Engineer

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Education

- **IP Paris - Université Dauphine PSL - M2 IREN Double degree** *Paris, France*
• Specialization in network industries and digital economy. *Sept. 2025 – Sept. 2026*
- **Télécom Paris Msc Engineering** *Paris, France*
• Specialization in Data Science (DS) and Markets, Organizations, Data, Strategies (MODS). *Sept. 2024 – Sept. 2026*
- **Universidad Nacional de Colombia Bsc Computer Engineering** *Bogotá, DC, Colombia*
• Full merit scholarship (top 1% nationwide, ICFES exam). *Mar. 2020 – Sept. 2026*

Work experience

- **AI Software architect, MODUO Ingénierie** *Sept. 2026*
- **AI Data Engineer Intern before IT Developer, MODUO Ingénierie** *Feb. 2026 – Sept. 2026*
 - Built automation tools in Python and VBA to streamline engineering workflows.
 - Designed a BIM data pipeline integrating Revit/IFC extraction, supplier pricing, and lifecycle assessment for automated cost and environmental analysis.
 - Developed RAG-based assistants for BIM documentation, improving knowledge retrieval and decision-making.
 - Built backend systems with Django and PostgreSQL to support internal applications.
- **Data Scientist, DNINFOA ,Universidad Nacional de Colombia** *May 2024 – Nov. 2024*
 - Led a team of 4 engineering students to design and implement a university policy Q&A chatbot.
 - Built a complete RAG pipeline with Ollama, Node.js/TypeScript backend, and RAG commander to improve response quality.
 - Delivered a functional prototype for administrative staff, reducing response time and improving operational efficiency.
- **Data Engineer Intern, Siigo S.A.S.** *Feb. 2024 – July 2024*
 - Built an internal RAG chatbot for the DataSmart platform with FastAPI backend and React UI, integrating Azure OpenAI with Azure Cognitive Search; deployed to QA and aligned with Siigo's brand guidelines.
 - Engineered a knowledge-base pipeline in Databricks using PySpark: extracted DB metadata and documents, applied per-document chunking with overlap, indexed via Cognitive Search, and automated ingestion to Blob Storage.
 - Created a RAG Commander to batch-test responses.
 - Added observability, persistence and delivered supporting data notebooks while collaborating with Azure FastTrack and the Data Science team under Scrum.
- **Data Manager, OPE - PRIG, Universidad Nacional de Colombia** *Sept. 2023 – Jan. 2024*
 - Designed, maintained ETL workflows to process research data efficiently, ensuring data quality, accessibility for the PRIG project.
 - Set up a cloud storage environment in AWS S3 integrated with Python scripts Excel for automated data ingestion and management.
- **Co-Founder, Co-Owner, Foundation Kawsay S.A.S** *Since Mar. 2022*
 - In charge of management data related with the foundation using Oracle DB, SQL, Excel, Python, the foundation's largest project was in 2023 in cooperation with the Túquerres City Hall, benefiting 200 families.

Academic and Professional Projects

- **Project, Credit Risk Modeling, Institut Polytechnique de Paris** *Sept. 2025 – Dec. 2025*
 - Built a loan default prediction pipeline on ~95k records using Logistic Regression and tree-based ensemble models with robust preprocessing and leakage control.
 - Achieved ~0.52 ROC-AUC (~50% predictive signal), identifying limitations in feature separability and improving evaluation strategies for imbalanced classification problems.
- **Software Engineer, TiendaUN, Universidad Nacional de Colombia** *Feb. 2024 – July 2024*
 - Designed and implemented a microservices platform with Node.js, TypeScript backend, using MongoDB Atlas, React web frontend.
 - Built an API Gateway, consumed a third-party SOAP service and exposed a partner-facing SOAP API
 - Security and access management with LDAP,JWT.Nginx reverse proxies (HTTPS) and performed QA-performance testing with JMeter, validating horizontal scaling improvements.
 - Deployed containerized services to Google Cloud with Docker, Kubernetes, load balancing, service replicas, and DB replication.

Skills

Cloud Service Models

- **IaaS:** Microsoft Azure, Google cloud, AWS
- **PaaS:** Databricks, Azure AI, Azure Cognitive Search, Azure OpenAI service, AWS S3, Cloud Run, Cloud Storage, Google Kubernetes Engine, Azure AI Search, Blob Storage,
- **SaaS:** Moduo Copil,Siigo Datasmart, TiendaUN.

Programming Languages

- Python, SQL, Java, JavaScript, TypeScript, R

Databases

- **SQL:** OracleDB, MySQL, PostgreSQL
- **NoSQL:** MongoDB.

DevOps & Tools: Docker, Kubernetes, CI/CD,Apache Tika, Selenium, JMeter, Git, Scrum, Databricks Workflows

Frameworks and Libraries

- **Data Engineering / AI:** Databricks Lakehouse, Delta Lake, Lakeflow (Jobs, Declarative Pipelines, Connect), PySpark, ETL, Data Processing, Data Modeling, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, RAG, Ollama, Qdrant.
- **Web:** FastAPI, Django, Node.js, SpringBoot, GraphQL, REST, SOAP, API Gateway
- **Protocols:** HTTP, HTTPS, LDAP, JWT, Proxies

Languages

SPANISH NATIVE | ENGLISH C1 | FRENCH C1